03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 5

(JCD 4.1-4.2)

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Previous lecture

- Ideal gas law:

$$P = \frac{\rho\, k\, T}{\mu\, m_u}$$

- Mean molecular weight:

$$\mu^{-1} = \sum_j X_j \frac{Z_j + 1}{A_j}$$

- Radiation pressure:

$$P_{\mathrm{rad}} = \frac{1}{3} a\, T^4$$

- Degenerate matter: $P_{\mathrm{deg,nr}} \propto \rho^{5/3}$ and $P_{\mathrm{deg,r}} \propto \rho^{4/3}$

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Outline

Mass equation

Hydrostatic equilibrium

Estimates of internal pressure and temperature

The importance of radiation pressure

Summary

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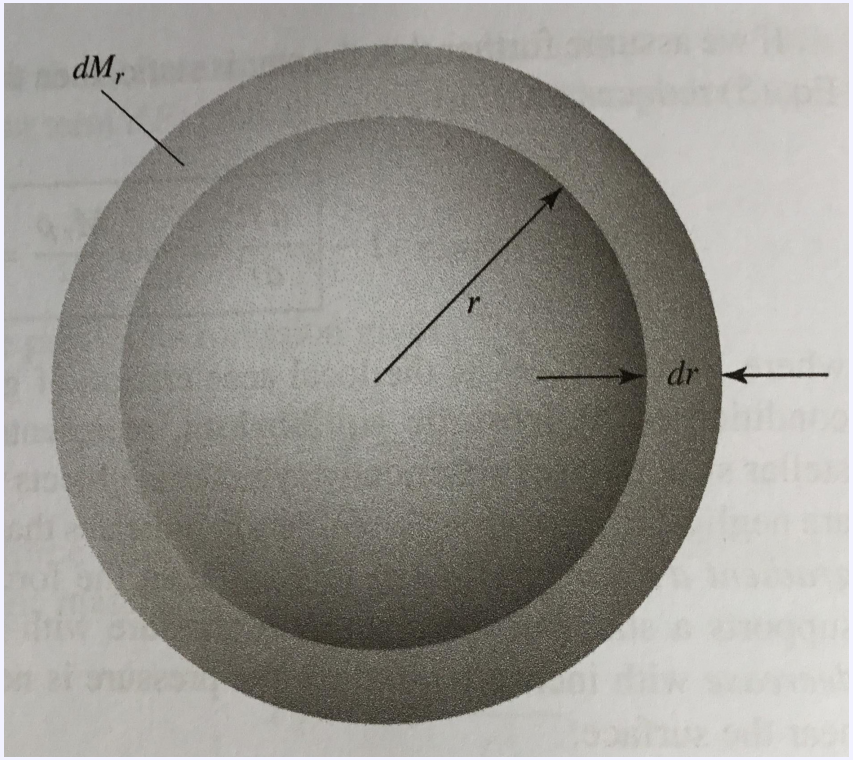
Mass equation



- Consider small spherical shell of width dr
- Within this small element in a star, properties are approximately constant (such as density)
- $dm(r) = dm = 4\pi r^2 \rho(r)\, dr$
- Mass interior to r is thus:

$$M(r) = 4\pi \int_0^r r^2 \rho(r)\, dr$$

$$\text{Mass equation: } \frac{dm(r)}{dr} = \frac{dm}{dr} = 4\pi r^2 \rho(r)$$

Figure credit: Carroll & Ostlie

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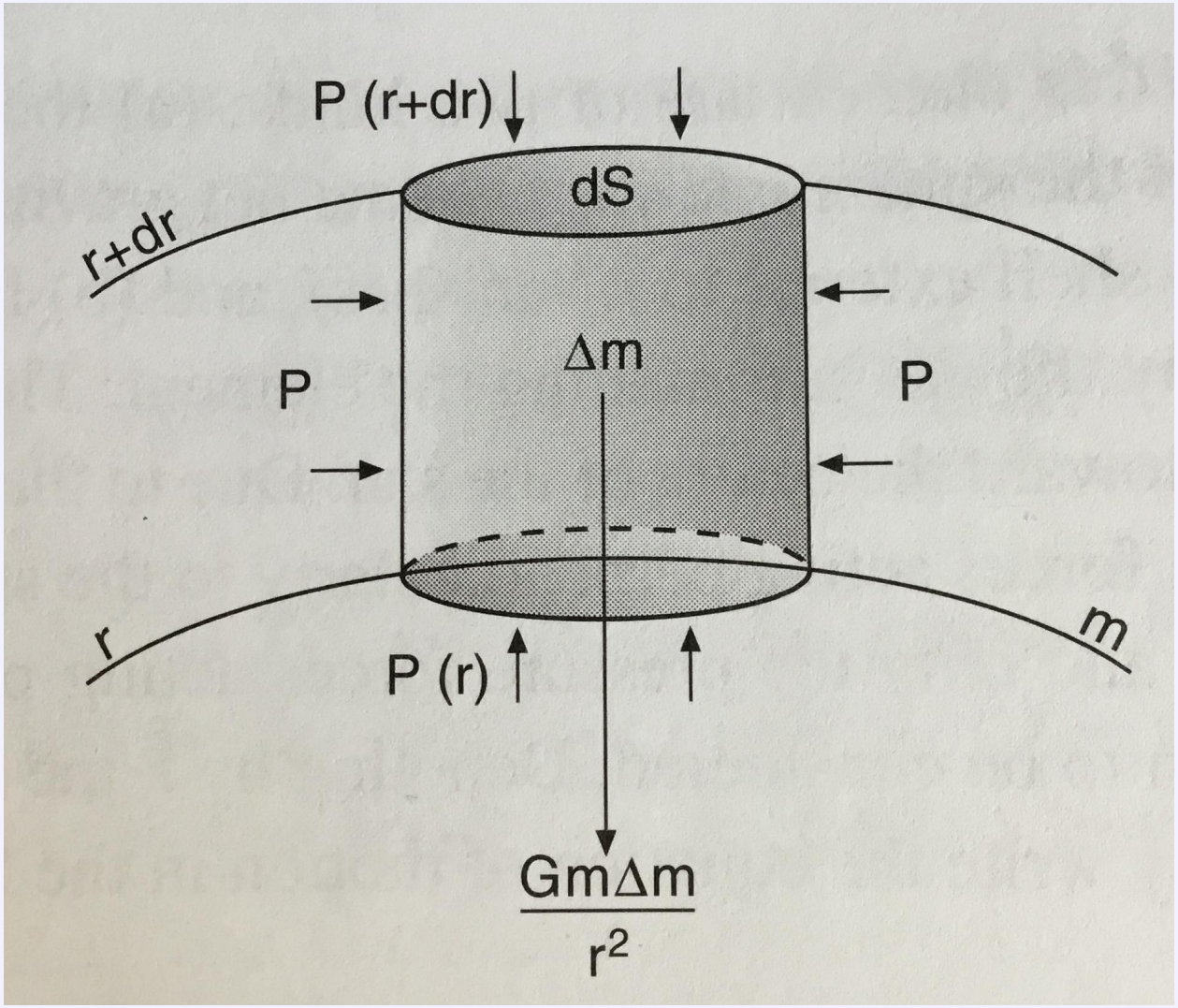
Stellar equilibrium

- Newton's second law: $F_{\text{net}} = \sum F_{\text{int}} = ma$
- Most stars are in a stable state
  $\Rightarrow$ Accelerations are negligible
  $\Rightarrow$ Net force equals zero
  $\Rightarrow$ There is a balance between all their internal forces
- Two types of forces acting in a star: gravitational and pressure force
- The balance of these forces is called **hydrostatic equilibrium**
- Remember lecture 1 - Hydrostatic equilibrium is a good assumption considering the involved timescales

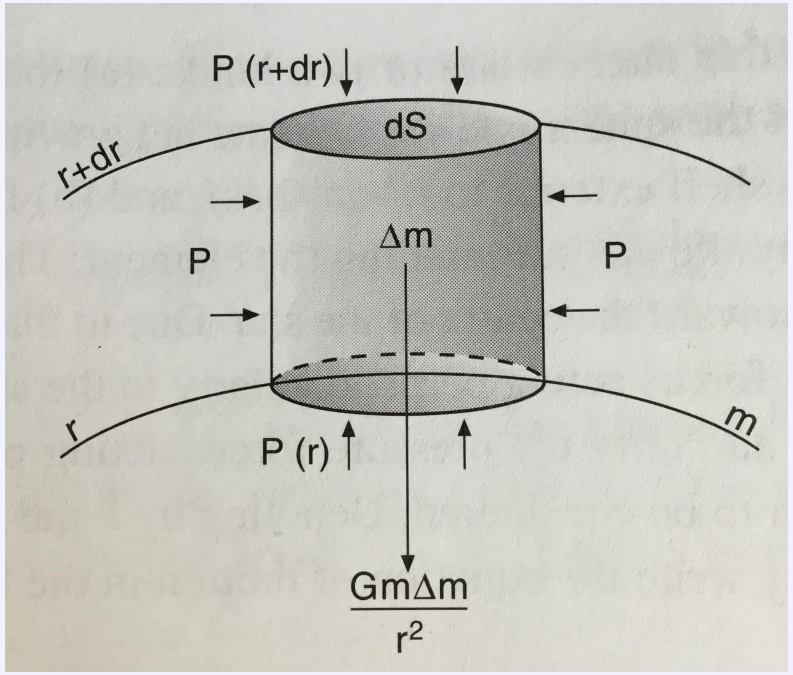


Consider a small cylindrical volume element in a star with mass

$$\Delta m = \rho \, dS \, dr$$

Figure credit: D. Prialnik

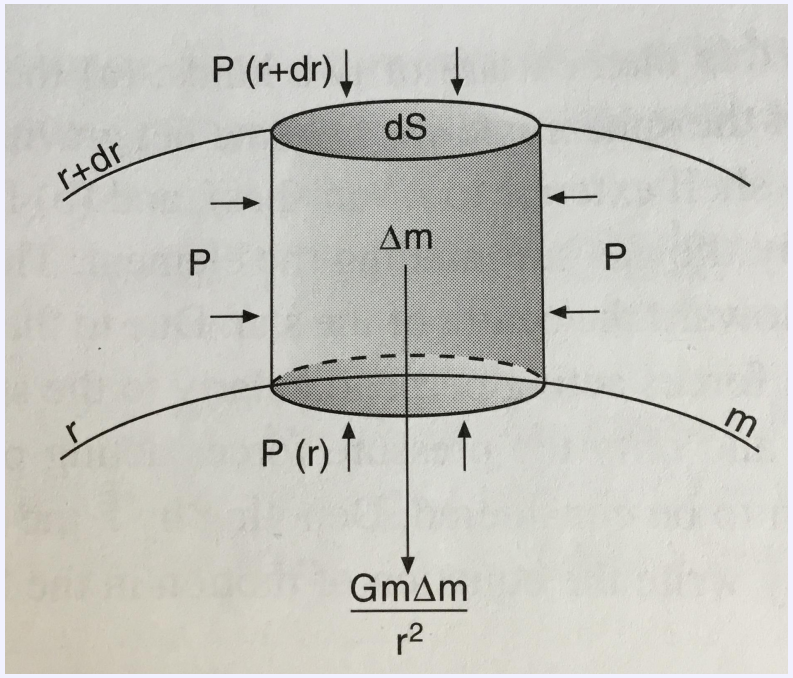
Gravitational force



- Only the mass interior to r plays a role.
  The outer layers have no effect.
- Mass interior to r: $M(r)$
- Gravitational force:

$$F_{\text{grav}} = \frac{GM(r)\Delta m}{r^2} = \frac{GM(r)\rho\, dS\, dr}{r^2}$$

Pressure force



- The sideward forces are balanced (spherical symmetry)
- Only (positive) upper and (negative) lower force remain
- Pressure force:

$$F_{\mathrm{p}} = P(r+dr)dS - P(r)dS = \frac{dP(r)}{dr}dr\,dS$$

Hydrostatic equilibrium

Sum of the forces is zero: $F_{\text{grav}} + F_{\text{p}} = 0$

$$\Downarrow$$

$$\frac{GM(r)\rho\, dS\, dr}{r^2} + \frac{dP(r)}{dr} dr\, dS = 0$$

$$\Downarrow$$

$$\underbrace{\frac{dP(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}}$$

$$\Downarrow$$

Equation of hydrostatic equilibrium

What does it mean?

$$\frac{dP(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

- For a star to be stable, there must be a pressure gradient to counteract the force of gravity
- RHS negative $\Rightarrow$ pressure decreases outward
- Note: you need to assume a density profile ($\rho(r)$) in order to solve
- Combined with the mass equation, you can show that

$$\frac{dP}{dm} = -\frac{GM(r)}{4\pi r^4}$$

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Estimating central pressure

$$\frac{dP(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

- Consider the full star
- $r \to R_*$, $M(r) \to M_*$
- Pressure gradient $\to -P_c/R_*$
- Density is approximated as mean stellar density

$$\frac{P_c}{R_*} \approx \frac{G\,M_*^2}{R_*^5} \Rightarrow P_c \approx \frac{G\,M_*^2}{R_*^4}$$

- For the Sun: $P_{c,\odot} \approx 10^{15}\,\mathrm{N\,m^{-2}} = 10^{16}\,\mathrm{dyn\,cm^{-2}}$
- Can use scaling relation again

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Estimating central temperature

- Use ideal gas law:

$$P = \frac{\rho k T}{\mu m_u}$$

- Remember central pressure estimate:

$$P_c \approx \frac{G M_*^2}{R_*^4}$$

$$\Rightarrow T_c = \frac{\mu_c m_u P_c}{k \rho_c} \approx \frac{G\, \mu_c m_u M_*}{k\, R_*}$$

- For the Sun: $T_{c,\odot} \approx 10^7$ K
- To use scaling relation, note that μ_c is star-specific and not a general constant

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The importance of radiation pressure

- Use estimates of T_c and P_c to find ratio of radiation to gas pressure
- Large caveat: Estimates were made assuming gas pressure was dominant!

$$\left(\frac{P_{rad}}{P}\right)_c \approx \frac{a\,T_c^4}{3\,P_c} \approx \frac{a\,G^3\mu_c^4\,m_u^4 M_*^2}{3\,k^4}$$

- For the Sun: $\left(\frac{P_{rad}}{P}\right)_{c,\odot} \approx 2 \cdot 10^{-3}$
- Ratio increases with mass (and mean molecular weight). Radiation pressure becomes more important for massive stars.

Summary

- Mass equation:

$$\frac{dm(r)}{dr} = \frac{dm}{dr} = 4\pi r^2 \rho(r)$$

- Equation of hydrostatic equilibrium:

$$\frac{dP(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

- Estimates of central pressure and temperature from basic physics principles not far of the modeled values
- Radiation pressure more important for massive stars

NEXT LECTURE: Virial theorem and simple models, polytropes